

PATENT

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

In re Application of	:	GOKARAJU et al.
	:	
For	:	ANTI-ADIPOGENIC
	:	COMPOSITIONS CONTAINING
	:	PIPER BETLE AND DOLICHOS
	:	BIFLORUS
	:	
Serial No.:	:	12/679,826
	:	
Filed	:	March 24, 2010
	:	
Art Unit	:	1655
	:	
Examiner	:	Amy Lynn CLARK
	:	
Att. Docket	:	LIX 3027
	:	
Confirmation No.	:	9829

RESPONSE TO RESTRICTION REQUIREMENT WITH AMENDMENT

Mail Stop Amendment
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Sir:

In response to the Office Action dated August 23, 2011, please amend the above-identified application as set forth below:

CLAIM AMENDMENTS begin on page 2 of this paper.

REMARKS/ARGUMENTS begin on page 12 of this paper.

CLAIM AMENDMENTS

This listing of claims will replace all prior versions and listings of claims in the application.

Listing of Claims

1-22. (Cancelled)

23. (Currently amended) Anti-adipogenic herbal compositions for inhibiting or preventing or ameliorating disease conditions associated with adipogenesis in mammals, comprising:

~~an effective amount from 33% to 67% by weight~~ of an extract or purified fraction derived from *Piper betle* and

an effective amount of an extract or purified fraction derived from *Dolichos biflorus*; optionally in combination with at least one biologically acceptable excipient.

24. (Currently amended) The anti-adipogenic compositions as claimed in claim 58 23, wherein the effective amount of an extract or purified fraction derived from *Piper betle* in the compositions varies from 1% to 99% by weight.

25. (Currently amended) The anti-adipogenic compositions as claimed in claim 58 23, wherein the composition comprises *Piper betle* extract in the range of 1% - 99% by weight and *Dolichos biflorus* extract in the range of from greater than 0% to 70% by weight.

26. (Currently amended) Anti-adipogenic herbal compositions ~~for inhibiting or preventing or ameliorating disease conditions associated with adipogenesis in mammals, comprising an effective amount of an extract or purified fraction derived from *Piper betle*, an effective amount of an extract or purified fraction derived from *Dolichos biflorus*, or a mixture thereof, optionally in combination with at least one biologically acceptable excipient according to claim~~
23,

wherein the composition comprises *Piper betle* extract in the range of 40% - 60% by weight and *Dolichos biflorus* extract in the range of 60 - 40% by weight.

27. (Previously Presented) Anti-adipogenic herbal compositions for inhibiting or preventing or ameliorating disease conditions associated with adipogenesis in mammals, comprising:

an effective amount of an extract or purified fraction derived from *Piper betle*;

optionally an effective amount of an extract or purified fraction derived from *Dolichos biflorus*;

optionally in combination with at least one biologically acceptable excipient;

wherein the said *Piper betle* extract further comprises one or more of the anti-adipogenic or anti-obesic agents selected from the group consisting of extracts or fractions derived from *Commiphora mukul*, *Boerhavia diffusa*, *Tribulus*

terrestris, and *Zingiber officinale*, and mixtures thereof and optionally contains a bio-enhancing agent or a bio-protecting agent.

28. (Previously Presented) The anti-adipogenic compositions as in claim 27, wherein the composition comprises extracts of *Piper betle*, *Commiphora mukul*, *Boerhavia diffusa*, *Tribulus terrestris*, *Zingiber officinale* along with said bio-enhancing agent, wherein said bio-enhancing agent comprises an extract of *Piper nigrum*.

29. (Previously Presented) The anti-adipogenic compositions as in claim 27, wherein the composition comprises *Piper betle* extract in the range of 10-40%, *Commiphora mukul* extract in the range of 10-40%, *Boerhavia diffusa* extract in the range of 10-40%, *Tribulus terrestris* extract in the range of 10-40%, *Zingiber officinale* extract in the range of 10 - 40% and said extract of *Piper nigrum* in the range of 10-40%.

30. (Previously Presented) The anti-adipogenic compositions as in claim 26, wherein the said compositions optionally comprise anti-adipogenic or anti-obese agents selected from the group consisting of an *Holoptelia integrifolia* extract, a *Garcinia cambogia* extract, a green tea extract, a green coffee bean extract, an eucalyptus plant extract, a bitter orange (*Citrus aurantium*) extract, Conjugated linoleic acid, a *Hoodia Gordonii* extract, an *Allium sativum* extract, chromium (III) complexes, DHEA, and 7-Keto-DHEA.

31. (Previously Presented) The anti-adipogenic compositions as claimed in claim 26, wherein said extracts or purified fractions or compositions are formulated optionally with biologically acceptable excipients into specific dosage forms selected from the group consisting of nutraceuticals, pharmaceuticals, and dietary forms.

32. (Previously Presented) The anti-adipogenic compositions as in claim 26, wherein said compositions further comprise an effective amount of at least one pharmaceutically or nutritionally or dietetically acceptable antioxidant, adaptogen, anti-inflammatory agent, anti-diabetic agent, bio-protectant, trace metals or mixture thereof to form a formulation.

33. (Previously Presented) The anti-adipogenic compositions as claimed in claim 26, wherein said composition is in a pharmaceutically or nutraceutically suitable form for oral administration, said pharmaceutically or nutraceutically suitable form being selected from the group consisting of tablets, soft capsules, hard capsules, pills, granules, powders, emulsions, suspensions, syrups, and pellets.

34. (Previously Presented) The anti-adipogenic compositions as claimed in claim 31, wherein the pharmaceutically or nutraceutically suitable form is suitable for parenteral administration selected from the group consisting of injections, drops, and suppositories.

35. (Previously Presented) The anti-adipogenic compositions as claimed in claim 26, wherein said compositions are in the form of a dietary formulation.

36. (Previously Presented) A pharmaceutically acceptable dosage form comprising a composition of claim 28 and a sufficient amount of a pharmaceutically acceptable excipient.

37. (Currently amended) A nutraceutical dosage form comprising a composition of claim 28, ~~optionally~~ further comprising a sufficient amount of a nutraceutically or dietetically acceptable excipient.

38. (Canceled)

39. (Previously Presented) The anti-adipogenic composition as claimed in claim 28, wherein said composition further comprises a pharmaceutically or nutraceutically or dietetically or veterinary acceptable additive, wherein the said additive is a binder, diluent, antioxidant or a lubricant.

40. (Previously Presented) The anti-adipogenic compositions as claimed in claim 23, wherein a source of the extract or purified fraction derived from *Piper betle* is a plant part selected from the group consisting of leaf, seed, trunk, root and combinations thereof; and

a source of the extract or purified fraction derived from *Dolichos biflorus* is a plant part selected from the group consisting of leaf, seed, trunk, root and combinations thereof.

41. (Previously Presented) The anti-adipogenic compositions as claimed in claim 23, wherein the extract or purified fraction derived from *Piper betle* and the extract or purified fraction derived from *Dolichos biflorus* are derived by extraction from plant parts with a solvent, wherein the solvent used for extraction of the plant parts is selected from the group consisting of methanol, ethanol, propanol, isopropanol, n-butanol, isobutanol, sec-butanol, tert-butanol, dimethyl ketone, methyl ethyl ketone, acetone, methyl isobutyl ketone, methyl acetate, ethyl acetate, butyl acetate, diethyl ether, water and mixtures thereof.

42. (Withdrawn) A method for inhibiting adipogenesis and fat accumulation in a mammal, comprising administering to a said mammal an effective amount of a composition as claimed in claim 28, as a dietary or nutraceutical or pharmaceutical composition.

43. (Withdrawn) A method for inhibiting or preventing or treating adipogenesis mediated diseases in a mammal comprising administering to a said mammal an effective amount of a composition as claimed in claim 28.

44. (Withdrawn) A weight controlling treatment regimen, comprising administering the anti-adipogenic compositions as claimed in claim 23, wherein the compositions are administered either before meals or as a meal replacement as a part of an overall weight control plan.

45. (Previously Presented) The anti-adipogenic compositions as claimed in claim 23, wherein the said compositions optionally comprise anti-adipogenic or anti-obese agents selected from the group consisting of an *Holoptelia integrifolia* extract, a *Garcinia cambogia* extract, a green tea extract, a green coffee bean extract, a eucalyptus plant extract, a bitter orange (*Citrus aurantium*) extract, Conjugated linoleic acid, a *Hoodia gordonii* extract, an *Allium sativum* extract, chromium (III) complexes, DHEA, and 7-Keto-DHEA.

46. (Previously Presented) The anti-adipogenic compositions as claimed in claim 23, wherein said extracts or purified fractions are formulated with biologically acceptable excipients into specific dosage forms, said dosage forms being selected from the group consisting of nutraceuticals, pharmaceuticals, or dietary forms.

47. (Previously Presented) The anti-adipogenic compositions as claimed in claim 32, wherein said bio-protectant is *Curcuma longa*.

48. (Previously Presented) The anti-adipogenic compositions as claimed in claim 26, wherein said compositions are in the form of a dietary formulation selected from the group consisting of chocolate bars, nutritional bars, creams, jams, gels, beverages selected from the group consisting of coffee, tea, a milk-containing beverage, a lactic acid bacteria-containing beverage, drops, candies, chewing gums, chocolates, gummy candies, yoghurt, ice cream, puddings, soft adzuki-bean jellies, jellies, and cookies.

49. (Previously Presented) A pharmaceutically acceptable dosage form comprising a composition of claim 23, further comprising a sufficient amount of a pharmaceutically acceptable excipient.

50. (Previously Presented) A nutraceutical dosage form comprising a composition of claim 23, further comprising a sufficient amount of a nutraceutically acceptable excipient.

51. (Previously Presented) A dietary dosage form comprising a composition of claim 23, further comprising a sufficient amount of a dietetically acceptable excipient.

52. (Withdrawn) A method for inhibiting or preventing or treating adipogenesis mediated diseases in a mammal comprising administering to a said mammal an effective amount of a composition as claimed in claim 23.

53. (Withdrawn) A method for inhibiting adipogenesis and fat accumulation in a mammal, comprising administering to a said mammal an effective amount of a composition as claimed in claim 23, as a dietary or nutraceutical or pharmaceutical composition.

54. (Withdrawn) A method for inhibiting or preventing or treating adipogenesis mediated diseases in a mammal comprising administering to a said mammal an effective amount of a composition as claimed in claim 26.

55. (Withdrawn) The method according to claim 54, wherein said adipogenesis mediated diseases are obesity, excess body weight, fat accumulation, or a combination thereof.

56. (Previously Presented) The composition comprising at least one component selected from the extract(s) and fraction(s) derived from *Piper betle* in combination with at least one component selected from the extract(s) and fraction(s) derived from *Dolichos biflorus*, optionally in combination with at least one biologically acceptable excipient.

57. (Previously Presented) The anti-adipogenic compositions as claimed in claim 23, wherein said compositions are in the form of a dietary formulation selected from the group consisting of chocolate bars, nutritional bars, creams, jams, gels, beverages selected from the group consisting of coffee, tea, a milk-containing

beverage, a lactic acid bacteria-containing beverage, drops, candies, chewing gums, chocolates, gummy candies, yoghurt, ice cream, puddings, soft adzuki-bean jellies, jellies, and cookies.

58. (New) An anti-adipogenic herbal composition for inhibiting disease conditions associated with adipogenesis in mammals, comprising:

an effective amount of an extract or purified fraction derived from *Piper betle*;

an effective amount of an extract or purified fraction derived from *Dolichos biflorus*; and

an optional biologically acceptable excipient;

wherein said extract or purified fraction derived from *Piper betle* and said extract or purified fraction derived from *Dolichos biflorus* are the only herbal ingredients present in said anti-adipogenic herbal composition.

59. (New) An anti-adipogenic herbal composition for inhibiting disease conditions associated with adipogenesis in mammals, comprising a first herbal component and an optional biologically acceptable excipient;

said first herbal component comprising an effective amount of an extract derived from *Piper betle* and an effective amount of an extract derived from *Dolichos biflorus*;

wherein said first herbal component is prepared by a process consisting essentially of mixing said extract derived from *Piper betle* and said extract derived from *Dolichos biflorus*.

60. (New) The anti-adipogenic compositions as claimed in claim 23, further comprising a second herbal component selected from the group consisting of an *Holoptelia integrifolia* extract, a *Garcinia cambogia* extract, a green tea extract, a green coffee bean extract, a eucalyptus plant extract, a bitter orange (*Citrus aurantium*) extract, a *Hoodia gordonii* extract, an *Allium sativum* extract, and mixtures thereof.

REMARKS/ARGUMENTS

By this amendment, claim 38 is canceled and claims 58-60 are added. Claims 23-26 and 37 are amended. No new matter has been added.

In response to the Restriction Requirement dated August 23, 2011, the Applicants elect Group I, claims 23-41, 45-51, 56 and 57, directed to the composition, with traverse.

The Applicants traverse the restriction requirement on the grounds that the composition and a therapeutic method of using the same are commonly classified, and can be searched together without placing an undue burden on the Office. We also traverse the restriction requirement on the grounds that claims 42-44 and 52-55 are linking claims which link a composition and a method of using the composition. According to MPEP § 809,

[w]here an application includes claims to distinct inventions as well as linking claims, restriction can nevertheless be required.

The linking claims must be examined with, and thus are considered part of, the invention elected. When all claims directed to the elected invention are allowable, should any linking claim be allowable, the restriction requirement between the linked inventions must be withdrawn.

Accordingly, linking claims 42-44 and 52-55 should be considered part of the invention directed to the composition. Therefore, the linking claims must be searched with the composition claims.

The Examiner has submitted arguments that claim 23 is anticipated by Bherata Bhaitajya Ratnekara (TKDL abstract), directed to a composition

comprising *Piper betle* and *Dolichos biflorus*. Independent claims 26 and 27 are not anticipated by the cited art. It is noted that the *Piper betle* in the cited art is present in an amount of no more than 1 part out of 5, i.e., 20%. Accordingly, claim 23 has been amended to recite that the claimed composition comprises between 33% and 67% *Piper betle*. Support for this amendment is found in Examples 5-7.

Additionally, new claim 58 recites herbal compositions comprising *Piper betle* and *Dolichos biflorus* as the only herbal compositions. Bherata Bhaitajya Ratnekara recites a composition comprising *Piper betle* and *Dolichos biflorus*, as well as *Ficus benghalensis* and *Butea monosperma*. Again, support for this claim is found in Examples 5-7.

Further, new claim 59 recites herbal compositions comprising a first herbal component containing *Piper betle* and *Dolichos biflorus*. The first herbal component is prepared by a process consisting essentially of mixing the extract derived from *Piper betle* and the extract derived from *Dolichos biflorus*. Support for this claim is found in Examples 5-7. Bherata Bhaitajya Ratnekara (TKDL abstract) describes a different composition prepared by heating the components in a crucible to produce an ash; the heating step is described in the abstract. The end product is a 'bhasma' or ash that is consumed. Since the heating step converts the extracts into an ash, the heating step materially changes the nature of the extracts in the first herbal component. Bherata Bhaitajya Ratnekara (TKDL abstract) provides no teaching or suggestion that the extract derived from *Piper betle* and the extract derived from *Dolichos biflorus* should be combined in the absence of a heating step.

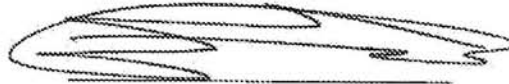
While we believe that the instant amendment places the application in condition for allowance, should the Examiner have any further comments or

suggestions, it is respectfully requested that the Examiner telephone the undersigned attorney in order to expeditiously resolve any outstanding issues.

In the event that the fees submitted prove to be insufficient in connection with the filing of this paper, please charge our Deposit Account Number 50-0578 and please credit any excess fees to such Deposit Account.

Respectfully submitted,
KRAMER & AMADO, P.C.

Date: September 16, 2011



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